

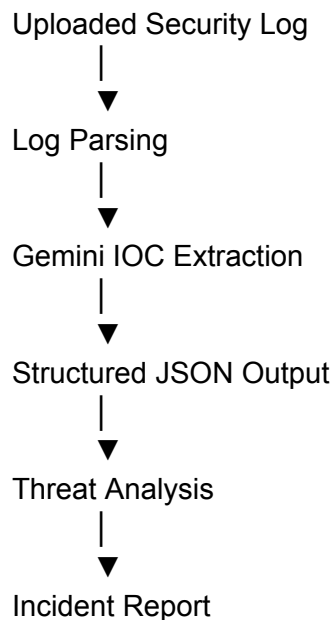
1. Title

Indicators of Compromise (IOC) Extraction

2. Objective

The IOC Extraction module automatically identifies Indicators of Compromise (IOCs) from uploaded security logs. These indicators are extracted using the Gemini Large Language Model and are used for threat analysis, MITRE ATT&CK mapping, risk assessment, and incident report generation.

3. IOC Extraction Workflow



4. IOC Categories Supported

This table is taken **directly from your `ioc_prompt.txt`**, so it matches your implementation exactly.

IOC Category	Description
IP Addresses	Extracts IPv4 and IPv6 addresses involved in the incident
Domains	Identifies suspicious or malicious domains
URLs	Detects URLs present in security logs
Email Addresses	Extracts sender or recipient email addresses
File Hashes	Detects MD5, SHA1 and SHA256 hashes
CVE IDs	Identifies vulnerability references in CVE format

5. JSON Output Schema

Your backend expects Gemini to return:

Field	Type
ips	Array
domains	Array
urls	Array
emails	Array
hashes	Array
cves	Array

6. Example (based on your screenshots)

Category	Example
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IP	192.168.1.45
IP	203.0.113.45
Domain	evil-command-control.example
Domain	login.microsoftonline.com
SHA256	8f4d5b3ac0fcb1d...
Email	unknown@evilmail.com
URL	https://evil-command-control.example

These examples match what is displayed in your application's IOC widget.

7. Why IOC Extraction Matters

Write:

- Identifies compromised hosts.
 - Detects malicious infrastructure.
 - Enables correlation across incidents.
 - Supports automated threat intelligence.
 - Improves incident response efficiency.
 - Assists in MITRE ATT&CK mapping.
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8. Integration within AI SOC Copilot

Describe the flow:

1. User uploads a security log.
2. The log is parsed into structured events.
3. Gemini extracts all available IOCs into structured JSON.
4. The extracted IOCs are forwarded to the Threat Analysis module.
5. The Threat Analysis module calculates severity, risk score, confidence, and MITRE ATT&CK mapping.
6. The final incident report includes the extracted IOCs for analyst review.

9. Conclusion

The IOC Extraction module provides standardized structured threat indicators that form the foundation of the AI SOC Copilot analysis pipeline. By extracting IP addresses, domains, URLs, email addresses, hashes, and CVE identifiers, the system enables comprehensive threat analysis and automated incident reporting.